Complete Eigen-Decomposition of a 2 × 2 Real Symmetric Matrix

Matrix chosen:

$$A = \begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$$

The matrix is real and symmetric because $A^T = A$.

1. Characteristic Equation

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 1 & 4-\lambda \end{bmatrix}$$

$$|A - \lambda I| = (4-\lambda)(4-\lambda) - (1)(1) = 0$$
$$(4-\lambda)^2 - 1 = 0$$
$$\lambda^2 - 8\lambda + 15 = 0$$
$$(\lambda - 5)(\lambda - 3) = 0$$

Eigenvalues: $\lambda_1 = 5$, $\lambda_2 = 3$. They are ordered from largest to smallest.

2. Eigenvector for $\lambda_1 = 5$

$$(A-5I)v = 0$$
$$\begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} [x,y]^T = 0$$
$$-x + y = 0 \quad \Rightarrow \quad y = x$$
Choose $v_1 = [1,1]^T$.
$||v_1|| = \sqrt{2}$, so $u_1 = (1/\sqrt{2})[1,1]^T$.

3. Eigenvector for $\lambda_2 = 3$

$$(A-3I)v = 0$$
$$\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} [x,y]^T = 0$$
$$x + y = 0 \quad \Rightarrow \quad y = -x$$
Choose $v_2 = [1,-1]^T$.
$||v_2|| = \sqrt{2}$, so $u_2 = (1/\sqrt{2})[1,-1]^T$.

4. Orthogonality and Matrix U

$$u_1 \cdot u_2 = (1/\sqrt{2})(1/\sqrt{2}) + (1/\sqrt{2})(-1/\sqrt{2})$$
$$= 1/2 - 1/2 = 0$$

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$
$$\Lambda = \begin{bmatrix} 5 & 0 \\ 0 & 3 \end{bmatrix}$$

Thus U is orthonormal, so $U^{-1} = U^T$.

5. Diagonalization Verification

$U^{-1}AU = U^{T}AU = [[5,0],[0,3]] = \Lambda$

Also, $U\Lambda U^{T} = [[4,1],[1,4]] = A$.

Conclusion: The matrix is diagonalizable by an orthogonal matrix U. Its ordered orthonormal eigenvectors are u_1 for $\lambda_1 = 5$ and u_2 for $\lambda_2 = 3$.